Chemistry-Aware Quantum Algorithms and Efficient Quantum Primitives

César Feniou

Sorbonne Université — LCT, Paris, France
Qubit Pharmaceuticals — Advanced Research Department

Outline

Context and motivation

High-level Spatial Resolution Challenge

Low-level efficient quantum primitives

Electronic Structure Theory

System. η electrons, M fixed nuclei $\{\mathbf{R}_\alpha, Z_\alpha\}$. Ground state solves $\hat{H}_{\text{el}}\Psi_0 = E_0\Psi_0$:

$$\hat{H}_{\text{el}} = \sum_{i=1}^{\eta}\left(-\tfrac{1}{2}\nabla_i^2 - \sum_\alpha \frac{Z_\alpha}{|\mathbf{r}_i - \mathbf{R}_\alpha|}\right) + \sum_{i<j}\frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \qquad \Psi_0 \in \bigwedge^{\eta} L^2\left(\mathbb{R}^3 \times \{\uparrow,\downarrow\}\right).$$

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- ▷ Single-electron space: $\mathcal{H}_1 = \mathrm{span}\{\varphi_m\}_{m=1}^{N} \longrightarrow \{|m\rangle\}_{m=1}^{N}, \quad \log N$ qubits.
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Exponents and prefactors improve with an **efficient Hamiltonian representation**.

Efficient Hamiltonian Representation

High-level — capture the physics.

- ▷ Basis set choice
- ▷ Discretisation & factorisation

Low-level — quantum primitives.

Circuit complexity of U_H

$\rightarrow$ size, depth, width

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$$\mathcal{C}_{\mathrm{QPE}} = \underbrace{\mathrm{cost}(W_H)}_{\substack{\text{sparsity \& structure of } \hat{H} \\ \text{naïvely: } \mathcal{O}(N^4) \text{ orbital integrals}}} \times \underbrace{\lambda}_{\substack{\ell_1\text{-norm} \\ \text{coeff. magnitudes}}} \times \underbrace{\varepsilon^{-1}}_{\substack{\text{target precision} \\ \varepsilon \sim 10^{-3}\,\mathrm{Ha}}}$$

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Resource estimates remain high — is there room for improvement?

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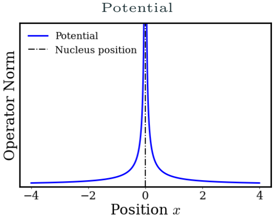
Coulomb potential induces localised wavefunction features:

- **Cusp at coalescence**: $\Psi \sim e^{-\alpha r_{ab}}$ as $r_{ab} \to 0$ (Kato cusp condition)
- **Smooth** behavior elsewhere; **exponential decay** as $r_{ab} \to \infty$.



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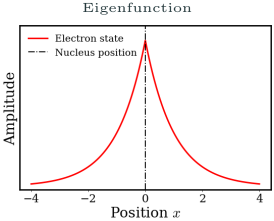

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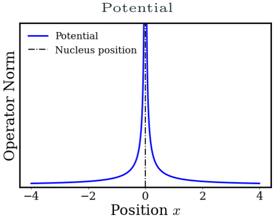

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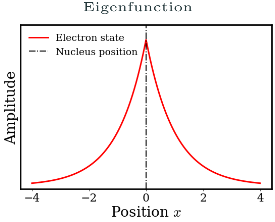

Difficulties for basis sets:

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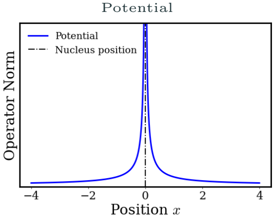

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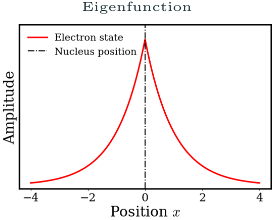

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Plane waves impose **uniform resolution**:
$\longrightarrow$ oversamples low-density regions to resolve cusps.

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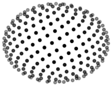
Idea. Repurpose multicenter **DFT integration grids** as basis:

- *Radial* — Becke mapping clusters points near nuclei: $r_i = -\alpha \ln(1 - u_i^\nu)$
- *Angular* — Lebedev quadrature preserves rotational symmetry.

Radial (Becke)



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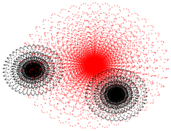

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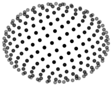

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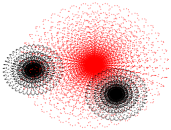

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DFT grids follow **molecular geometry** and **electron density**
⇒ natural **adaptive real-space** discretisation basis.

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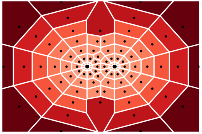

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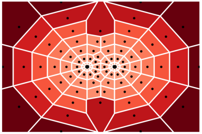

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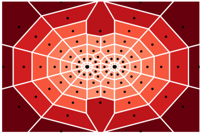

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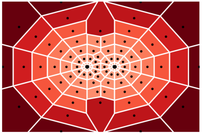

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- **Coulomb** terms are diagonal $\mathbf{U}_i = \sum_\alpha \frac{Z_\alpha}{\|\mathbf{r}_i - \mathbf{R}_\alpha\|}$, $\quad \mathbf{W}_{i,j} = \frac{1}{\|\mathbf{r}_i - \mathbf{r}_j\|}$

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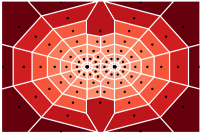
Nontrivial differential operator formulation (overlapping, non-tensorised grids):

- **Voronoi partition** of $\{\mathbf{r}_m\}$ defines unambiguous neighbors.
- **Finite-volume** discretisation of differential operators.

Laplacian $$\mathbf{L}_{mn} = \begin{cases} -\dfrac{1}{v_m} \displaystyle\sum_{k \in \Lambda(m)} \dfrac{\sigma_{mk}}{\|\mathbf{r}_m - \mathbf{r}_k\|}, & m = n, \\ \dfrac{1}{v_m} \dfrac{\sigma_{mn}}{\|\mathbf{r}_m - \mathbf{r}_n\|}, & n \in \Lambda(m), \end{cases}$$



Voronoi cells.

- $\mathcal{O}(1)$-sparse; neighbor count **does not scale** in fixed dimension
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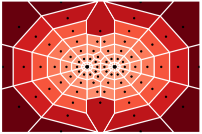

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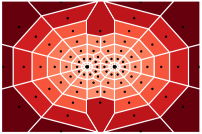

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Hamiltonian in an **adaptive grid basis** suitable for QPE, but at what cost?

Operator Norm vs. Spatial Resolution

$$\|\text{Laplacian}\| \sim h_{\min}^{-2}, \qquad \|\text{Coulombian}\| \sim h_{\min}^{-1}, \qquad \lambda_H = \mathcal{O}\left(\frac{\eta}{h_{\min}^2} + \frac{\eta^2}{h_{\min}}\right)$$

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Becke-type maps $r_i = -\alpha \ln(1 - u_i^\nu)$: compress $h_{\min}$ exponentially in N.

Uniform resolution	$\longleftrightarrow$	**Adaptive resolution**
slow CBS convergence		fast CBS convergence
$\lambda \sim \mathcal{O}(N^{2/3})$		$\lambda \sim 2^N$

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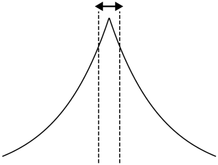
$$h_{\min}^{\varepsilon} = \max\left\{h > 0 \;:\; \left|E_{\mathrm{FCI}}^{(h)} - E_{\mathrm{FCI}}^{\mathrm{CBS}}\right| \leq \varepsilon\right\}$$



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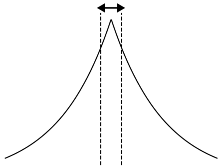

- ▷ Prefactors gains over uniform grids to be assessed on real problems.
- ▷ Is the **norm/resolution trade-off** fundamentally unavoidable?
- ▷ How about **electron–electron cusps**?

Transcorrelated Hamiltonian

Transcorrelated Hamiltonian

Encode coulombic cusps analytically via a similarity transform:

$$\hat{H}_{TC} := e^{-\tau}\,\hat{H}\,e^{\tau} = \hat{H} + [\hat{H},\tau] + \tfrac{1}{2}[[\hat{H},\tau],\tau],$$

with symmetric $\tau = \sum_i \Big(\sum_\alpha g_{\mu_{ne}}(x_{i\alpha}) + \sum_{j>i} h_{\mu_{ee}}(x_{ij}) \Big)$

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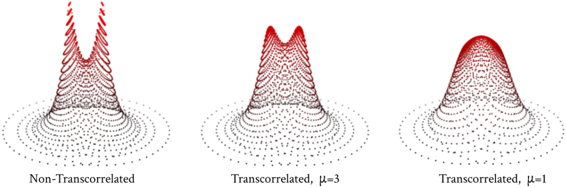

$\tilde{H}_{\mathrm{TC}}\,|\psi_0\rangle = E_0\,|\psi_0\rangle$ solved for H_2^+ ground state on a 2D adaptive Voronoi grid.

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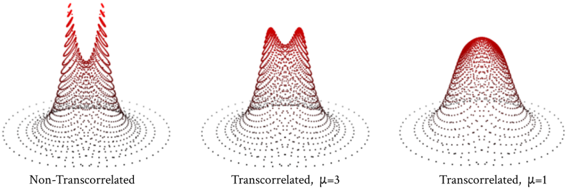

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$$\tilde{\mathbf{H}} = \sum_i \tilde{\mathbf{T}}_i + \sum_{i<j} \tilde{\mathbf{W}}_{ij} + \sum_{i<j<k} \tilde{\mathbf{B}}_{ijk}.$$

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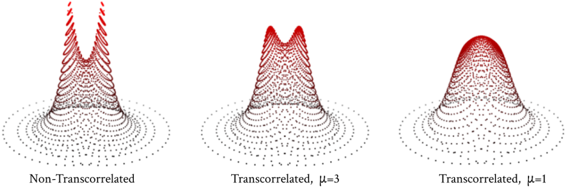

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Smooth eigenfunctions $\Rightarrow$ less critical spatial resolution

Cost: $\hat{H}_{\mathrm{TC}}$ is **non-Hermitian** and acquires **three-body terms**.

Numerics[3]

Non-Hermitian requires **Generalised Quantum Eigenvalue Estimation**.[1]

1 Low & Su (2024) — *Quantum Eigenvalue Processing*, FOCS.

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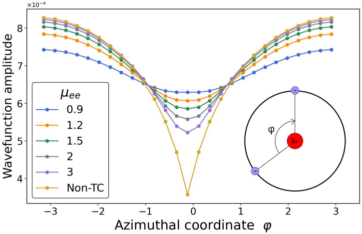
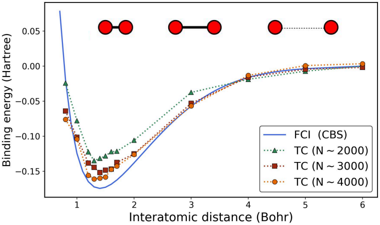
Electron–electron cusp profile



H_2 dissociation profile
4k grid points → matrix size 16 million



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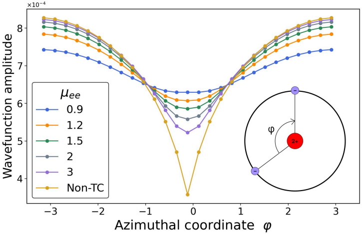

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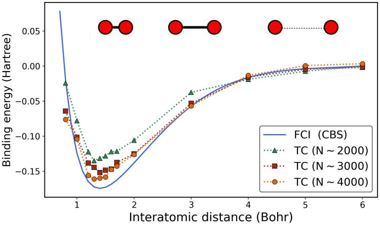

Correct cusp behaviour, size consistency, and systematic improvability with N.

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Summary: Chemistry-Aware Spatial Discretisation

Two compatible levers to improve spatial resolution in first-quantised QPE:

Adaptive grids

Cluster points in relevant regions
$\Rightarrow$ better $\varepsilon_{\text{basis}}$-to-$N$ ratio.

Transcorrelated Hamiltonian

Encode cusps analytically
$\Rightarrow$ smoother eigenfunctions

Drawbacks:

- Suboptimal grid and TC parameters yet — results remain qualitative.
- Breaking the fully-tensorised Hamiltonian structure increases block-encoding cost.
- TC Hamiltonian introduces three-body terms $\Rightarrow$ naïve $\mathcal{O}(\eta^3)$ overhead.

Can improved quantum primitives make these Hamiltonian representations competitive?

Outline

Context and motivation

High-level Spatial Resolution Challenge

Low-level efficient quantum primitives

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Quantum algorithms rely on subroutines decomposed into primitive gates:

QFT, multi-controlled gates, arithmetic circuits, diagonal operators, ...

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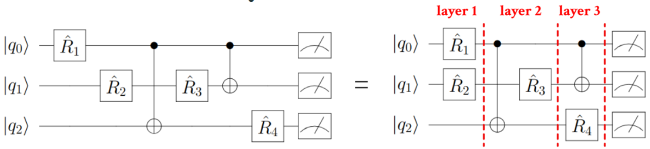
The decomposition sets the **quantum resource cost**:

Quantum circuit



Width: number of qubits **Size**: total number of gates **Depth**: number of layers

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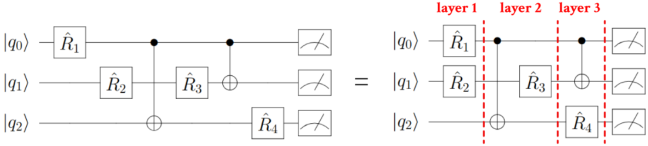

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Improving these primitives directly propagate to the cost of algorithms.

Contributions Recap

1. Polynomial state preparation.

$$|p\rangle \propto \sum_x p(x)|x\rangle, \quad p(x) = \sum_{k=0}^{d} a_k x^k$$

Prior art	$\mathcal{O}(nd)$ depth	$\mathcal{O}(\log n)$ ancilla
Our work[1]	$\mathcal{O}(\mathbf{d}\log \mathbf{n})$ **depth**	$\mathcal{O}(n)$ ancilla

2. Exact-one oracle. Flag Hamming-weight-one states $|1/2^k\rangle \equiv |\underbrace{0\cdots0}_{k-1}1\underbrace{0\cdots0}_{n-k}\rangle$:

$$\mathcal{O}_n|x\rangle|0\rangle = \begin{cases} |x\rangle|1\rangle & x = |1/2^k\rangle \\ |x\rangle|0\rangle & \text{else} \end{cases}$$

Prior art	$\mathcal{O}(\log n)$ depth	$\mathcal{O}(n)$ ancilla
Our work[2]	$\mathcal{O}(\log n)$ depth	**2 ancilla**

3. Multi-controlled NOT gate ($n-$qubit Toffoli).

Flip target iff all n controls are on.

Prior art	$\mathcal{O}(n)$ depth	any ancilla regime
Our work[2]	$\mathcal{O}(\log^{\mathbf{3}} \mathbf{n})$ **depth**	any ancilla regime
Following work[3]	$\mathcal{O}(\log \mathbf{n})$ **depth**	2 ancilla

[1] Claudon, Lucas, Piquemal, Feniou, Zylberman, *arXiv:2603.16527* (2026).

[2] Claudon, Zylberman, Feniou, Peruzzo, Piquemal, *Nat. Commun.* **15**, 5886 (2024).

[3] Khattar, Gidney, *Quantum* **9**, 1752 (2025).

Quantum State Preparation of Polynomials

Problem. Prepare $|p\rangle \propto \sum_x p(x)|x\rangle$ for $p(x) = \sum_{k=0}^{d} a_k x^k$.

Strategy.[1] Build a diagonal polynomial operator then apply to $|+\rangle = H^{\otimes n}|0\rangle$:

$$\underbrace{\alpha I + \sum_{k=1}^{n} \frac{Z_k}{2^k}}_{\text{Pauli decomp. of affine op.}} \propto \sum_x x|x\rangle\langle x| \xrightarrow{\text{GQET}} \underbrace{\sum_x p(x)|x\rangle\langle x|}_{\text{promote to degree } d} \xrightarrow{|+\rangle} \sum_x p(x)|x\rangle.$$

log depth via LCU

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	Depth	**Width**	**Size**	**Succ. prob.**
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Prior art (parallel LCU)	$\mathcal{O}(d \log n)$	$\mathcal{O}(e^n)$	$\mathcal{O}(dn)$	$\mathcal{O}(1)$
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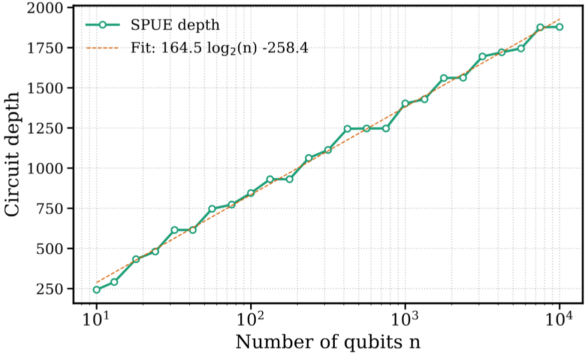
- **Polynomial approximations** are everywhere in computational physics.
- **Diagonal polynomial operators** useful beyond state preparation (e.g. harmonic potential in real space).

[1] B. Claudon, A. Lucas, J.-P. Piquemal, C. Feniou, J. Zylberman, arXiv:2603.16527 (2026).

Numerics — Circuit Depth Scaling

Circuit depth for affine diagonal operator.

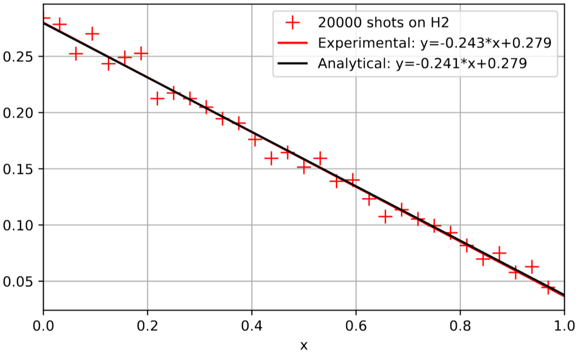
The $\mathcal{O}(d \log n)$ scaling is confirmed numerically, and prefactors are fitted.



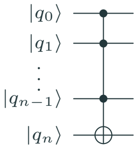
Numerics — Hardware Demonstration

Execution on Quantinuum H2 — 5 qubits + 9 ancilla.

The affine state preparation is compiled and run on the Quantinuum H2 trapped-ion processor.



The Multi-Controlled NOT Gate



Conditionally-clean ancilla enables recursive decomposition:

$$n \to \sqrt{n} \to \sqrt{\sqrt{n}} \to \cdots \quad \Rightarrow \quad D(n) = \Theta(\log^3 n).$$

Table 1 | Circuit depth of ancilla-free methods

Method	Circuit depth
Gidney[18]	$494n - 1413$
Silva (ϵ)[28]	$64n + 1645$ if $(\epsilon = 10^{-7})$
Proposition 2 (ϵ)	$\lceil \log(\pi/\epsilon) \rceil (86 \log(n)^3 - 2564)$

Table 2 | Circuit depth of borrowed ancillae methods

Method	Circuit depth	Borrowed
Barenco 1[15]	$48n - 148$	1
Barenco $n-2$[15]	$24n - 43$	$n-2$
Proposition 1	$43 \log(n)^3 - 1287$	1

Table 3 | Circuit depth of zeroed ancillae methods

Method	Circuit depth	Zeroed
Barenco 1[15]	$36n - 111$	1
Barenco $n-2$[15]	$12n - 12$	$n-2$
Proposition 1	$27 \log(n)^3 - 808$	1
Proposition 3	$27 \log(n/\lceil m/2 \rceil)^3 +$	$2 \le m \le n$
	$16 \lceil \log(\lfloor m/2 \rfloor) \rceil - 808$	
He[22]	$16 \lceil \log(n) \rceil + 12$	n

Exponential depth improvement:

$$\mathcal{O}(n) \;\Rightarrow\; \Theta(\log^3 n)$$

in all ancilla regimes.
Application: SELECT operator.

[1] Claudon, Zylberman, Feniou, Peruzzo, Piquemal, *Nat. Commun.* **15**, 5886 (2024)

Conclusion

Conclusion

High-level. Chemistry-aware Hamiltonian representation.

- **Adaptive grids**
- **Transcorrelated Hamiltonian**
- Pseudopotentials
- Discontinuous Galerkin...

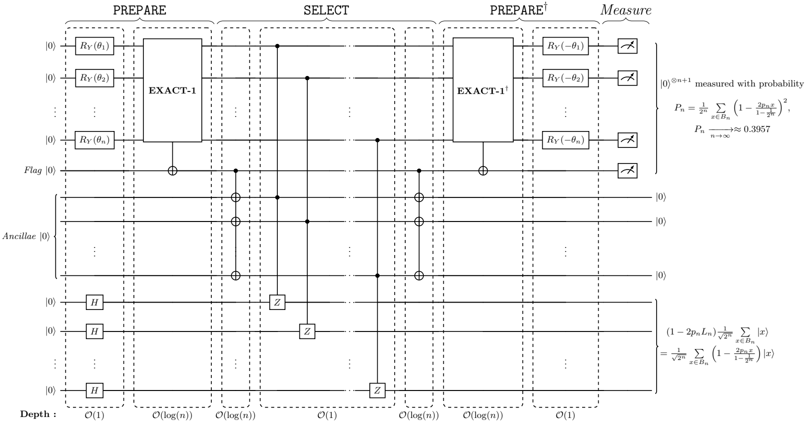
$\updownarrow$ **co-design** $\updownarrow$

Low-level. Efficient quantum primitives.

- **Multi-control NOT**
- **Polynomial state prep**
- **Exact-one oracle**
- Exact-k oracle?
- Log-depth QFT?
- Hyperbolic diagonal op?

[1] Feniou, Cherfan, Zylberman, Claudon, Piquemal, Giner, *arXiv:2507.20583.*
[2] Claudon, Zylberman, Feniou, Peruzzo, Piquemal, *Nat. Commun.* **15**, 5886.
[3] Claudon, Lucas, Piquemal, Feniou, Zylberman, *arXiv:2603.16527.*

Logarithmic-Depth Polynomial State Preparation



Block-encoding U_{aff}, with $\theta_k = 2\arcsin\left(\sqrt{\frac{1}{1+2^k}}\right)$:

`PREPARE` $\rightarrow$ unary `SELECT` on the one-hot subspace $\rightarrow$ `PREPARE`† $\rightarrow$ postselection.